Effects of soil drainage types on macrofaunal community composition and density

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Abstract
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Keywords
soil macrofauna, soil drainage, Paracou, Amazon forest, soil ecology

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1. Introduction

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2. Material and Methods

2.1 Study Site

Geographic Location

This study was conducted in Amazonian primary tropical forest at the Paracou Research Station (5°16 38 N, 52°55 38 W). The average air temperature is 25.7 °C ± 0.1 °C (from 2004 to 2014). Average annual rainfall is 3,102 mm ± 70 mm, with heavy rains during the wet season from December to July (between 300 and 500 mm per month) and a long dry season from mid-August to mid-November (< 100 mm per month). In Paracou, a permanent forest dynamic experiment was established in 1984. The site comprises 15 plots measuring 250 × 250 m^2 (6.25 ha) and one plot measuring 500 × 500 m^2 (25 ha). Our experiment was carried-out in two of the unlogged control plots (see Figure 1): Plot 1 (P1), installed in 1984, and Plot 15 (P15) in 1990 (Gourlet-Fleury et al., 2004).

Geological and Pedological Characteristics of Paracou

The geological substratum of Paracou consists primarily of Precambrian schists intruded by pegmatite veins, belonging to the metamorphic Bonidoro Series (Choubert, 1974). The site exhibits a relatively low elevation (5 to 50 m a.s.l.), dissected by deep drainage gullies flowing into the Sinnamary River (Gourlet-Fleury et al., 2004). The predominant soils are nutrient-poor Acrisols (sesquioxide, kaolinite and gibbsite) (Driessen et al., 2001; FAO et al., 1998), characterized by a shallow depth limited by a more or less transformed loamy saprolite (Gourlet-Fleury et al., 2004).

Soil distribution and drainage varies strongly in Paracou (Roelens, 2007; Soucémarianadin, 2004; Weigel, 2006), influenced by the topographic gradient, with Plot 1 comprised of 45 % Superficial Lateral Drainage (SLD), 35 % Surface Hydromorphy (SH), 15 % Downhill System (DhS), and 5 % Deep Vertical Drainage/altered phase (Alt/DVD),

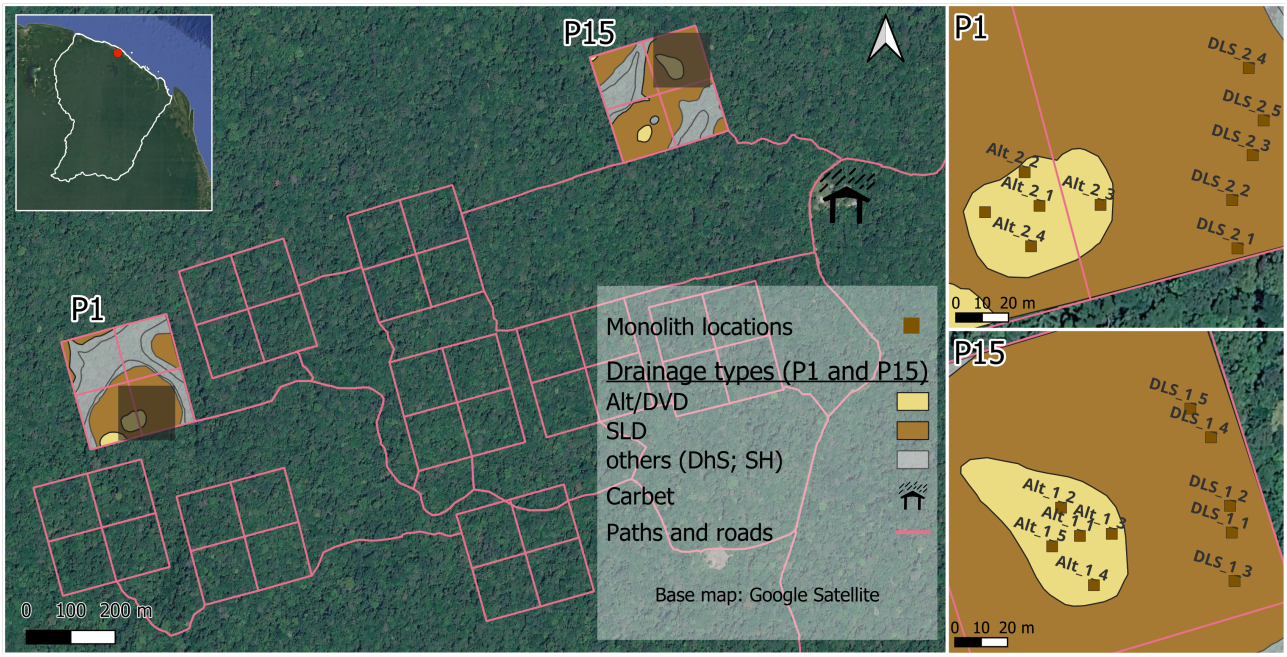


Figure 1. Overview map of Paracou with the sampling locations. Location of right panels shown by grey squares. White lines on the right panels represent altitude isolines. Upper left corner panel shows location of Paracou in French Guiana.

according to Soucémarianadin (2004). Meanwhile, the soils of P15 are comprised of 60 % SLD, 20 % DhS, 15 % SH and 5 % Alt/DVD (Soucémarianadin, 2004).

Alt/DVD are a type of well-drained soils (hypoferralic acrisols) located on flat hilltops or plateaus and feature a thick ferralitic cover that lets rainwater percolate freely in depth. The altered phase (Alt) begins where red, muscovite-rich alloterite appears above 1.2 m depth, signaling initial soil thinning (Sabatier et al., 1997). Plateau soils contain higher levels of carbon, nitrogen, and aluminum (Allié et al., 2015), support greater root biomass and soil respiration (Epron et al., 2006), and host nearly half of the local tree species in Paracou (Allié et al., 2015).

2.2 Sampling and Identification

To evaluate how soil type influences macrofaunal communities, two contrasting ferralitic soil (DVS/Alt and SLD) types were compared, previously mapped in Paracou (Roelens, 2007; Soucémarianadin, 2004; Weigel, 2006).

%%% WHAT ABOUT SLD soils

Soil Type Verification

Prior to macrofauna sampling, field checks were carried out to confirm the pedological classification of our sampling sites. Auger drillings were performed to assess soil color and texture (Boulet et al., 1979; Sabatier et al., 1997). Additionally, silt and clay contents were confirmed through granulometric analysis conducted by the company SADEF. Furthermore, saturated hydraulic conductivity was determined in situ using the simplified Beerkan infiltration method (Lassabatère et al., 2006).

Marofaunal Sampling

Five macrofauna samples were collected in each soil type (Alt/DVD and SLD) per plot (P1 and P15) using the Tropical Soil Biology and Fertility (TSBF) method by Anderson & Ingram (1994).

To account for local topography, Alt/DVD (hilltop) samples were arranged in a cross-pattern, whereas SLD (mid-slope) samples were aligned along a transect following the slope. All sampling points were spaced at least 10 m apart to reduce spatial dependance of macrofauna samples (see Figure 1 for spatial representation).

Following the TSBF method (Anderson & Ingram, 1994), a 25 × 25 × 10 cm metal frame was placed on the ground at for each sample (called monolith). First, the litter (L) was quickly removed, trying to avoid fauna to escape, and collected in a labeled bag. Next, the frame is pressed 10 cm deep into the soil to isolate a soil block. This process was performed twice to collect the soil layers of 0 to 10 cm (10) and 10 to 20 cm (20) depth which were also collected in separate bags.

Sample Sorting and Identification

Samples were manually sorted at the field station and litter samples (L) processed first to avoid macrofauna from escaping. Soil was sorted by transferring around 200 g of substrate into a container. Each portion was observed for an average of 2 minutes by an observer. Around 5 observers were involved in the sorting of each bag to minimize bias. Macrofauna was collected using a brush and stored in 90 % ethanol, separated according to monolith and layer of soil. Subsequently, the organisms were identified on a coarse taxonomic level (refer to the data file in Section 7 for details), with ants (Formicidae) and termites (Isoptera) treated separately from other Hymenoptera and Blattodea.

2.3 Statistical analysis

All analyses was performed using R version 4.6.1 (R Core Team, 2026). Plots and data processing relied largely on the tidyverse package (Wickham et al., 2019; Wickham, 2023).

Explorative Analysis

%%% TEXT

Abundances (Generalized Linear Mixed Models)

Effects of soil drainage type and soil layers on abundances were analyzed using Generalized Linear Mixed Models (GLMMs), fitted with the glmmTMB package (M. Brooks et al., 2026; M. E. Brooks et al., 2017). To account spatial structure, monoliths were included as a random effect. Since count data was treated, Poisson and negative binomial (NB2) were tested. Model selection relied on Akaike's Information Criterion (AIC), Likelihood-ratio testing (LRT) and model diagnostics using DHARMa residuals (Hartig, 2026). Refer to Section 7 for detailed model selection and diagnostic plots.

Type II Wald χ^2 tests from the package car (Fox et al., 2026; Fox & Weisberg, 2019) were used to evaluate overall predictor effects. Estimated marginal means and Tukey pairwise comparisons were generated using emmeans (Lenth & Piaskowski, 2026).

Community Composition (PERMANOVA)

%%% Jason: will write this section later vegan (Oksanen et al., 2026): adonis2, betadisper, permutest, envfit permute; mention zero removal (Simpson, 2026): permutation schemes Estimated marginal means and Tukey pairwise comparisons emmeans (Lenth & Piaskowski, 2026).

2.4 Spatial Representation

Monolith locations were reconstructed using R (R Core Team, 2026) and the sf package (Pebesma, 2018; Pebesma & Bivand, 2023). Coordinates were obtained based on distance and azimuth in relation to reference trees and their coordinates sourced from Paracou inventory data (Derroire et al., 2026a, 2026b). QGIS (QGIS Development Team, 2024) was used for spatial representation.

3. Results

3.1 General

%%% TEXT explorative

Densities by soil type and layer Figure 2

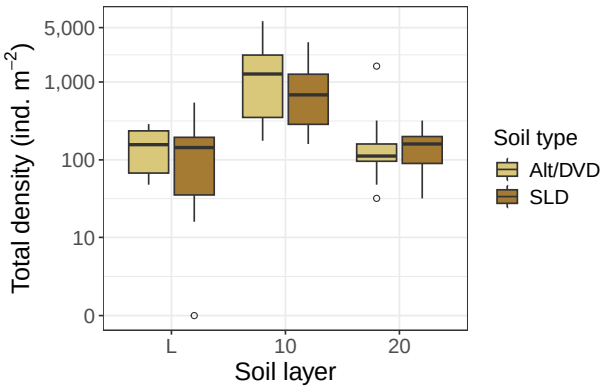


Figure 2. Boxplot with total macrofauna densities by soil type and layer. y-axis is on a pseudo-logarithmic scale.

%%% TEXT Relative composition of community abundances Figure 3

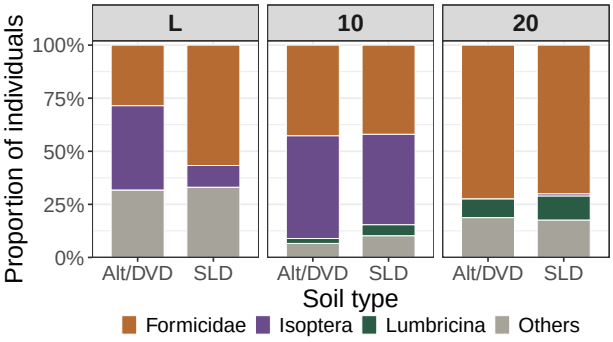


Figure 3. Relative composition of macrofaunal community abundances by soil type and layer.

%%% TEXT Explorative ordination PCA Figure 4

3.2 GLMM

Effects of soil type and layer were tested on global, ant, termite and earthworm abundances. All retained models described below are listed in Table 1.

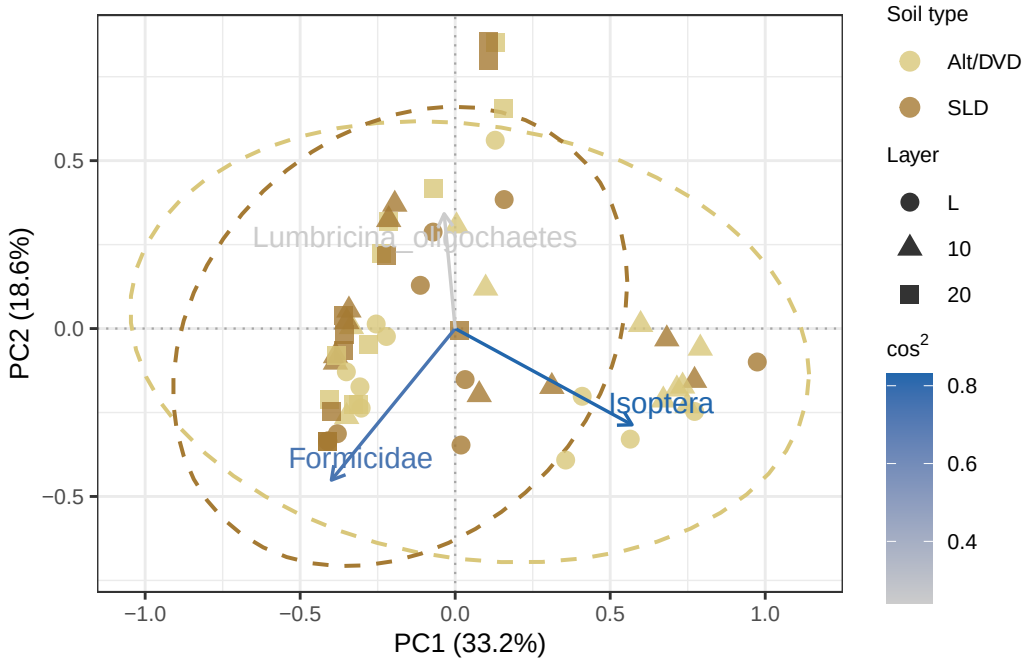


Figure 4. PCA ordination based on Hellinger-transformed community matrix. First two dimensions are presented. Only taxa with $\cos^2 > 0.1$ are presented.

Global Abundance

Type II Wald χ^2 tests revealed a highly significant effect of soil layer on total soil macrofauna abundance ($\chi^2 = 106.85$, df $= 2$, $p < 0.001$). The effect of soil type was not significant ($\chi^2 = 0.52$, df $= 1$, $p = 0.469$). Estimated random-effect variance (monolith) was 0.437 (SD $= 0.661$), indicating variation in baseline macrofauna abundance among monoliths.

Pairwise comparisons with Tukey adjustment (Table 2) showed that macrofauna abundance was significantly higher in layer 10 than in layers L (7.42 times) and 20 (6.18 times), while no significant difference was found between layers L and 20 ($p = 0.714$).

Formicidae

Type II Wald χ^2 tests revealed a significant effect of soil layer on ant abundance ($\chi^2 = 28.00$, df $= 2$, $p < 0.001$). The effect of soil type was not significant ($\chi^2 = 0.48$, df $= 1$, $p = 0.486$). The estimated random-effect variance for monolith was 0.655 (SD $= 0.810$), indicating variation in baseline ant abundance among monoliths. Quantile deviation was detected in the DHARMa residuals for the presented model. Specifically, upper quantile deviated significantly and thus relatively high-abundance observations may not be adequately captured by the model. Since the diagnostics did not indicate deviations regarding dispersion, zero-inflation, uniformity as well outliers, and the model has the best AIC support, it was retained, regardless.

Pairwise comparisons with Tukey adjustment (Table 2) showed that ant abundance was significantly higher in layer 10 than in layer L (5.05 times) and layer 20 (3.27 times), while no significant difference wasfound between layers L and 20 ($p = 0.3995$).

Isoptera

Type II Wald χ^2 tests revealed a highly significant effect of soil layer on termite abundance ($\chi^2 = 36.56$, df $= 2$, $p < 0.001$).The effect of soil type was not significant ($\chi^2 = 0.89$, df $= 1$, $p = 0.347$). The estimated random-effect variance for monolith was effectively zero (8.22×10^{-9}; SD $= 0.00009$), indicating virtually no remaining variation in baseline termite abundance among monoliths after accounting for the fixed effects.

Pairwise comparisons with Tukey adjustment (Table 2) showed that termite abundance was significantly higher in layer 10 than in layer L (18.08 times) and layer 20 (745.48 times; estimated richness for 20 nearing zero). Abundance was also significantly higher in layer L than in layer 20 (41.23 times; all p 0.0073).

Lumbricina Oligochaetes

Type II Wald χ^2 tests indicated a significant effect of soil layer on earthworm abundance ($\chi^2 =$

Table 1. Summary of retained GLMMs for global, ants (Formicidae), termite (Isoptera) and earthworm (Lumbricina oligochaetes) abundance.

Response	Distribution	Predictor	Chi-square	df	p-value
Total macrofauna	Negative binomial	soil_type	0.52	1	0.46933
		layer	106.85	2	<0.0001***
Ants (Formicidae)	Negative binomial	soil_type	0.48	1	0.48633
		layer	28.00	2	<0.0001***
Termites (Isoptera)	Negative binomial	soil_type	0.89	1	0.34664
		layer	36.56	2	<0.0001***
Earthworms (Lumbricina)	Negative binomial	soil_type	0.35	1	0.55692
		layer	12.69	2	0.00176**

Type II Wald Chi-square tests. All models included soil type and soil layer as fixed effects and monolith ID as a random intercept. Significance levels: * p < 0.05; ** p < 0.01; *** p < 0.001

Table 2. Estimated marginal means and Tukey pairwise comparisons of abundance by layer for total macrofauna, ants (Formicidae) and termites (Isoptera) using emmeans.

Taxon	Comparison	Abundance ratio	z-ratio	Adjusted p-value
Total macrofauna	L - 10	0.13	-9.12	<0.0001***
	L - 20	0.83	-0.78	0.713689
	10 - 20	6.18	8.34	<0.0001***
Ants (Formicidae)	L - 10	0.20	-4.96	<0.0001***
	L - 20	0.65	-1.29	0.399482
	10 - 20	3.27	3.84	0.000363***
Termites (Isoptera)	L - 10	0.06	-4.06	0.000145***
	L - 20	41.23	3.01	0.007262**
	10 - 20	745.48	5.42	<0.0001***

Abundance ratios are exponentiated log-scale contrasts. Significance levels: * p < 0.05; ** p < 0.01; *** p < 0.001. P-values adjusted using the Tukey method within each taxon.

12.69), df = 2, p = 0.0018),whereas soil type had no significant effect ($\chi^2 = 0.35$), df = 1, p = 0.557). Pairwise comparisons showed significantly higher earthworm abundance in layer 10 compared to layer 20 (2.31 times higher). No significant differences were detected between layer L and the other layers. However, no earthworms were recorded in layer L, resulting in unstable model estimates and unbounded confidence intervals for this layer. The Estimated random-effect variance for monoliths was 0.510 (SD = 0.714)

3.3 PERMANOVA

PERMANOVA did not show a significant difference in macrofaunal community composition between soil types (F = 1.68, $R^2 = 0.085$, p = 0.139). The test for homogeneity of multivariate dispersion was also non-significant (F = 0.11, p = 0.742), indicating no detectable difference in within-group dispersion between soil types. Layers were aggregated per monolith to account for spatial dependence.

PERMANOVA revealed a significant effect of soil layer on macrofauna community composition (Bray–Curtis dissimilarity: F = 5.46, $R^2 = 0.163$,p = 0.0001). Soil layer explained around 16.3% of variation in the community composition, indicating that macrofauna community composition differed significantly among layers. Permutation was restricted within monolith to account for spatial dependence of layers. Homogeneity of multivariate dispersion was assessed using a permutation test based on betadisper. No significant differences in dispersion were detected for soil layers (F = 0.75, p = 0.321).

Table 3. Pairwise PERMANOVA for community composition comparisons between layers.

Comparison	F-statistic	R^2	Adjusted p-value
L - 10	6.05	0.141	0.0003***
L - 20	4.55	0.110	0.0003***
10 - 20	5.76	0.132	0.0021**

Significance levels: * p < 0.05; ** p < 0.01; *** p < 0.001; p-values adjusted using Bonferroni method

Pairwise PERMANOVA suggests significant differences in macrofauna community composition for all soil layers (Table 3). Significant differences were observed between L and 10 (F = 6.05, R^2 = 0.141, adjusted p = 0.0003), L and 20 (F = 4.55, $R^2 = 0.110$, adjusted p = 0.0003), and 10 and 20 (F = 5.76, $R^2 = 0.132$, adjusted p = 0.0012), using Bonferroni correction.

%%% results shown, TEXT Figure 5

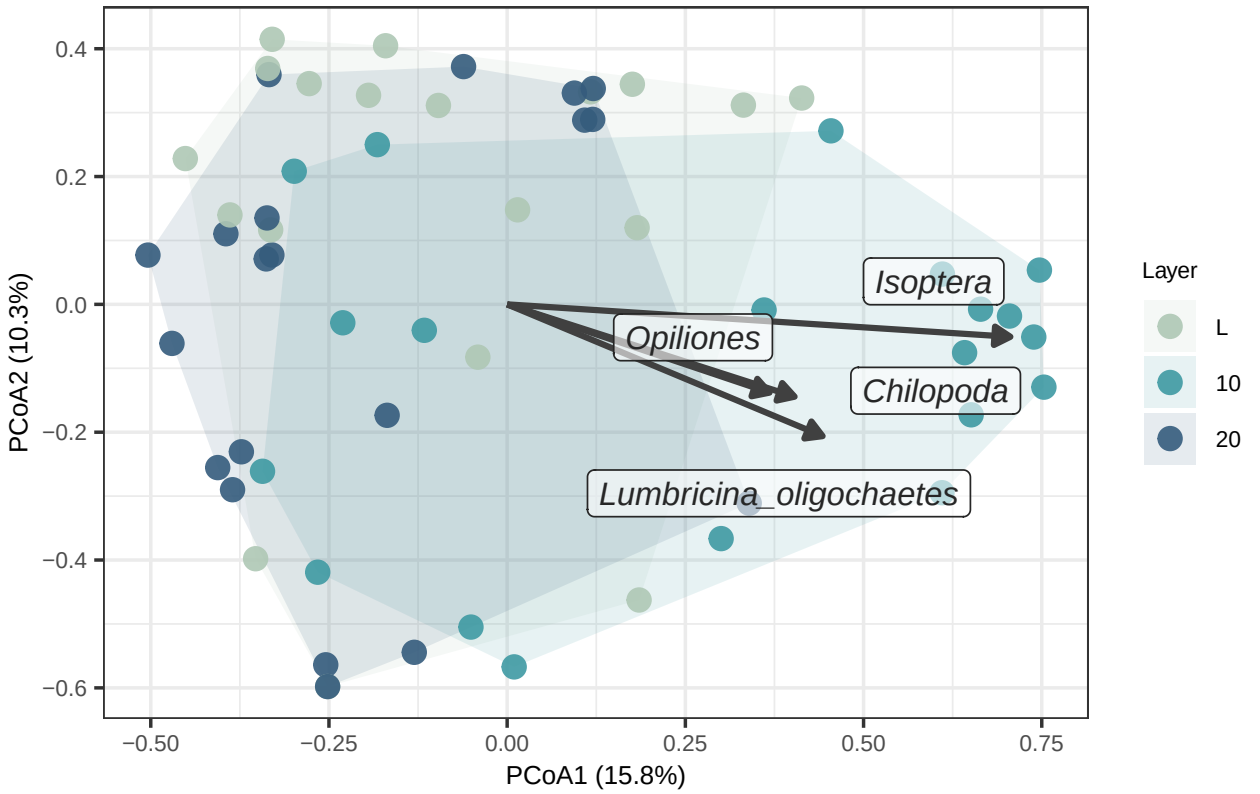


Figure 5. First two components of the Bray-Curtis PCoA representing 26.1% of the variation in macrofaunal community composition. Husks indicate soil layers. Taxa singificantly associated with the ordination are presented (arrows).

4. Discussion

%%% TEXT - no differences in densities or community composition for soil types (argue about same vegetation cover etc.) - clear differences in both for layers (connect to literature knowledge)

5. Conclusion

%%% TEXT

6. Acknowledgements

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7. Supplementary Material

Supplementary material, including detailed statistical analysis, scripts, data, QGIS layers, additional software attribution as well as the Quarto file used to produce this article, is available from https://github.com/jasonberger2002/FTH_2026_soil_macrofauna.

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